



Seismological study of the central Ecuadorian margin: Evidence of upper plate deformation

Nicole Bethoux^{a,*}, Monica Segovia^b, Viviana Alvarez^{a,b}, Jean-Yves Collot^a, Philippe Charvis^a, Audrey Gailler^a, Tony Monfret^a

^a Université de Nice, UMR GéoAzur, Observatoire de la Côte d'Azur, BP 48, 06235 Villefranche sur Mer, France

^b Instituto Geofísica-Escuela Politécnica Nacional, Av. Ladrón de Guevara E11-253 y 12 de Octubre, Quito, Ecuador

ARTICLE INFO

Article history:

Received 12 November 2009

Accepted 22 August 2010

Keywords:

Seismicity

Deformation

Ecuadorian margin

North Andean block

Palabras clave:

Sismicidad

deformación

margen ecuatoriano

Bloque Norandino

ABSTRACT

A seismic study of a segment of the convergent margin of Ecuador is presented. During the SISTEUR campaign a network of 24 Ocean Bottom Seismometers (OBS) was deployed on the Carnegie Ridge, one line along the main axes of the ridge and two lines across the strike of the edge of the ridge, during one month. This marine network was complemented with a land network of 20 stations distributed in two lines: one parallel to the margin and the other perpendicular to it.

The seismic event recorded by these networks, were located using different crustal models defined from the wide-angle seismic data modeling. Relative location techniques were used to improve earthquake locations. Seismogram waveform modeling allowed us to constrain hypocentral location for events farther than ~50 km from the network. This modeling also provided additional information to constrain the focal mechanisms of these events. The upper limit of the Interplate Seismogenic Zone (ISZ) is estimated to be at a 10 km depth in the region. The background seismic activity of the upper plate provided new insights:

1) A seismic cluster that reaches the base of the overriding plate is linked to the Jipijapa-Portoviejo fault. The reactivation of this Quaternary fault is confirmed by focal mechanisms that provide rupture planes parallel to its superficial projection (N10°–N25°).

2) The focal mechanisms presented in this study are compatible with a homogeneous regional stress field corresponding to an E–W to ESE–WNW compression and an NNE–SSW extension. The presence of strike-slip deformation, with a reverse component, corresponds to the NNE escape of the North Andean Block. Normal faulting accommodating this movement suggests that this part of the North Andean Block cannot be considered as a rigid block.

© 2010 Elsevier Ltd. All rights reserved.

R E S U M E N

Se presenta un estudio sísmico del margen convergente de Ecuador. Durante la campaña SISTEUR se instaló una red de 24 sismómetros marinos (OBS) en la Cordillera de Carnegie, una línea a lo largo del eje de la cordillera y dos líneas paralelas al margen convergente, durante un mes. Este trabajo fue complementado con la instalación de una red de 20 estaciones en el margen, distribuidas en dos líneas: una paralela al margen y otra perpendicular a éste.

Los sismos registrados por estas dos redes fueron localizados usando diferentes modelos de velocidad definidos con la modelación de datos sísmicos de gran ángulo. Técnicas de localización relativa se utilizaron para mejorar las ubicaciones. El modelamiento de las formas de onda permitió constreñir la localización hipocentral de los eventos ubicados más allá de 50 km de la red. Este modelamiento también proveyó información adicional para constreñir los mecanismos focales de estos eventos. La profundidad del límite de la zona sismogénica interplacas en esta zona se estima en los 10 km. El registro de la sismicidad de fondo proporcionó nuevos indicios:

* Corresponding author.

E-mail address: nbethoux@geoazur.obs-vlfr.fr (N. Bethoux).

1) La presencia de actividad microsísmica que llega hasta la base de la placa superior está relacionada con la falla Jipijapa-Portoviejo. La reactivación de esta falla Cuaternaria se confirma con los mecanismos focales que proporcionan planos de ruptura paralelos a su proyección superficial (N10°–N25°).

2) Los mecanismos focales obtenidos son compatibles con un campo de esfuerzos regional homogéneo con una dirección de compresión E–O a ESE–ONO y una extensión NNE–SSO. La presencia de fallas de rumbo con componente inversa, responde al escape del Bloque Norandino en la dirección NNE. El fallamiento normal que acomoda este movimiento sugiere que esta parte del Bloque Norandino no se puede considerar como un bloque rígido.

© 2010 Elsevier Ltd. All rights reserved.

1. Introduction

The Ecuador–Colombian margin encompasses two seismically and tectonically contrasted segments (Collot et al., 2002): a northern segment (Latitude: 3.5°N–0.5°S) that underwent great historical earthquakes, such as 1906, $M = 8.7$ and a southern segment (Latitude: 0.5°S–2.0°S) without such seismic activity.

The northern zone is located just north of the Carnegie ridge (Fig. 1) and its subduction under the Andean margin seems to act as the limit of two these zones. However, the area where this ridge is subducted is subject to regular seismic activity with events with magnitude up to 6. The last crisis occurred in 2005 near Manta (Fig. 1). The swarm had four events of magnitude greater than 6, 11 events with $5 < M_L < 6$ and 470 events with $5 < M_L < 6$ (Vaca et al., 2010). On the northern flank of the ridge, in the Bahía region, the seismicity catalogues contain several events of magnitudes higher than 7. The major event was the Bahía earthquake of magnitude $M_w = 7.1$ in 1998 (Segovia, 2001).

However, the seismicity of the Ecuadorian margin is poorly known. World catalogues deal only with events of magnitude greater than 5 whereas the lower magnitude seismicity is usually detected and located by Ecuadorian permanent network, maintained by the Geophysical Institute of Quito (RENSIG). This network is mainly concentrated in the Andean cordillera around the active volcanoes (Fig. 1). Concerning the coastal or offshore seismicity, uncertainties in hypocentral locations are consequently important and a significant part of the small to intermediate seismicity ($2 \leq M_W \leq 4$), which is likely to contain key information about the active deformation processes, is not recorded. Thus, the seismicity pattern and stress field of the Ecuadorian margin are poorly defined.

In this context, this short seismic experiment is useful to derive new information about the active deformation of the central part of the Ecuadorian margin. Studying the seismicity at the transition zone between these two different segments of the margin is of great interest for both seismic risk and geodynamic concerns.

This study deals with data collected during the marine seismic SISTER campaign, which was performed in 2000 (Collot et al., 2002, 2004; Graindorge et al., 2004; Sage et al., 2006) to image the interplate seismogenic zone. The study presented here focuses on the Ecuadorian margin around Latitude 1.4°S. A network of 24 Ocean Bottom Seismometers (OBS) was deployed across both the inner and outer subduction trench walls extending westward onto the Carnegie ridge (Fig. 2). They were deployed along a principal axis perpendicular to the trench and along two lines parallel to the margin. This marine network was complemented by a land network of 20 stations distributed along two lines: one parallel to the margin and the other perpendicular to it. This combined land-sea network recorded the shots produced by air guns. This network configuration was chosen for a 2D wide-angle study (Graindorge et al., 2004; Gailler et al., 2007) and 3D modeling (Gailler, 2005).

The lack of permanent seismological stations and the poor knowledge of the local seismicity led us to develop a new methodology to employ the recorded data. To balance the poor azimuthal coverage of the network we took advantage of the knowledge of

the velocity structure from the ridge up to the coastal region, obtained from the SISTER experiment. Using waveform modeling we were also able to constrain some hypocenters and determine focal mechanisms. The purpose of this study was first to evaluate the upper limit of the Interplate Seismogenic Zone (ISZ) in this part of the margin, and second, to improve the knowledge of the upper plate seismicity in the central coastal zone. Based on the relocation of the micro-seismic events and the computation of focal solutions, tectonic implications of these results are proposed.

2. Geodynamical and structural setting

Northwestern corner of South America has a complex geodynamic evolution due to the interaction between the Nazca, South America and Caribbean Plates and North Andean Block (NAB) (Fig. 1). The Nazca plate, which is being subducted under the Andean margin derives from the fragmentation of the Farallon plate ~23 My ago (Herron, 1972; Handshumacher, 1976; Hey, 1977; Minster and Jordan, 1978; Hammerickx et al., 1980; Wortel and Cloetingh, 1981; Wortel, 1984). During the Neogene times, interaction between the Galápagos hotspot and the Nazca Plate generated the NE-trending Cocos ridge and the E-trending Carnegie ridge (Pennington, 1981; Sallares and Charvis, 2003). The Malpelo ridge is thought to be the former continuation of the Cocos ridge, drifted away by the dextral strike-slip motion along the Panama fracture zone (Longsdale and Klitgord, 1978). These ridges are characterized by irregular topography, with important bathymetric variations and a thickened oceanic crust which can reach 19° km (Sallares et al., 2005; Gailler et al., 2007). The NAB consists of oceanic terrains that were accreted to the Andean margin during compressive periods in Late Paleocene–Early Eocene times (Jaillard et al., 1997). Ecuador's coastal region is therefore underlain by oceanic type crust, known as the Piñón formation which is overlain by a Neogene sedimentary basin with a thick fill called the “Manabi basin” (Fig. 1).

The convergence between the Nazca and South America plate is ~58 mm/yr (Trenkamp et al., 2002) trending towards N82°E (DeMets et al., 1990). As a consequence of this collision the NAB is escaping towards the NE along the Dolores-Guayaquil Megashear (Fig. 1). This movement results in the opening of the Gulf of Guayaquil and also the formation of Quaternary NW–SE normal faulting in the region (Benítez, 1995; Daly, 1989; Deniaud, 2000; Dumont et al., 2005; Witt et al., 2006). Nevertheless, for other authors, the escape of the NAB would be accommodated by The Major Dextral System (Pennington, 1981; Kellog and Bonini, 1982; Mann and Burke, 1984; Toussaint and Restrepo, 1987; Mann and Corrigan, 1990; Soulas et al., 1991) that initiates at Guayaquil Gulf, continues in Pallatanga (Western Cordillera) and jumps to the Eastern Cordillera with the Chingual-La Sofía fault. In agreement with this scheme the Dolores-Guayaquil Megashear corresponding to the suture between allocthonous lands of the Western Cordillera and the Interandean Valley–Eastern Cordillera does not control the present tectonic activity (Soulas et al., 1991).

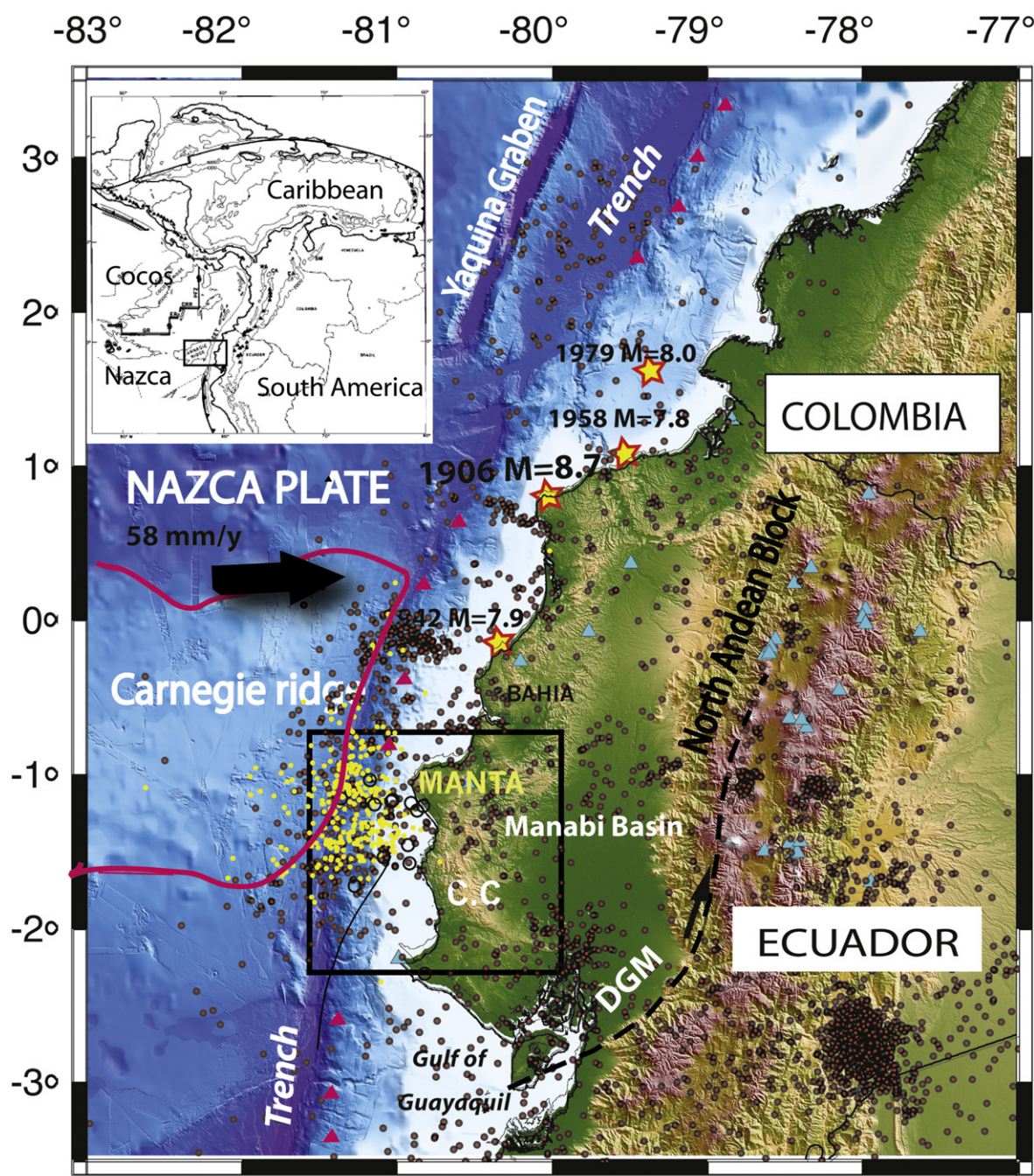


Fig. 1. Geodynamical sketch of the North Andean block and Ecuadorian margin. Yellow stars represent the epicenters of the four great earthquakes which occurred during the 20th century (Collot et al., 2004). The Nazca plate motion vector is from Trenkamp et al. (2002). DGM is the Dolores-Guayaquil Megashear. The Manabi basin and the Coastal Cordillera (CC) are indicated. Seismicity from Rensig catalogue (1994–2004) is superimposed (red circles), whereas yellow circles and empty circles ($M_w > 5$) correspond to the 2005 seismic crisis of Manta. The seismicological stations available in 2000 are indicated by blue triangles. The limits of the tectonic plates modified from Pennington (1981). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

The main Quaternary faults recognized in the field or detected by aerial photographs are shown in the Ecuadorian Neotectonic Map (Egüez et al., 2003). One part of this map is shown in Fig. 3. We use this document in order to better understand the seismicity pattern we obtained. A new geomorphologic study was recently carried out using a DEM (Reyes, 2008; Reyes et al., 2010), which studied the profiles and incision of the rivers combined with a morpho-structural analysis; the study concluded that the Coastal Cordillera is segmented into blocks each of which had its own

period of uplift. Two fault systems seemed to guide the evolution of the coastal Cordillera, the Jipijapa system and the Jama system (Fig. 3A). The upper plate seismicity of the margin does not appear to be associated with these geological structures recognized in the field (Guillier et al., 2001; Segovia and Alvarado, 2010). Here, we show that the temporary seismic network, located around the Jipijapa fault system allows us to demonstrate the seismic activity of this inherited structure. The so-called Portoviejo-Jipijapa fault has been defined from the geological studies of the Ecuadorian

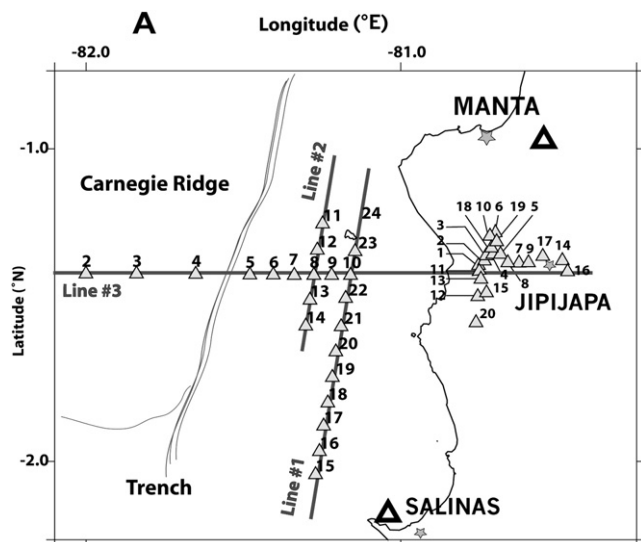


Fig. 2. Marine and land temporary seismic networks in antenna configuration (white triangles). The two permanent RENSIG stations are reported as black triangles. The three seismic lines are superimposed.

margin (Daly, 1989; Deniaud, 2000; Egüez et al., 2003) and projected down to 20 km depth on the basis of seismic petroleum profiles. The Jipijapa fault limits the Manabi basin to the west and is described in the literature as a strike-slip duplex (Fig. 3B).

3. Data

The network configuration for the experiment is shown in Fig. 2. Because it was installed mainly for a wide-angle experiment, its geometry involves an E-trending, 200 km-long antenna with a main axis that included 9 OBS and 13 land stations deployed from the trench up to the foothills of the Coastal Cordillera, at Latitude $\sim 1.4^\circ\text{S}$. Three perpendicular axes, parallel to the margin, complemented the network with 9 OBS and 7 land stations. The land seismic network included 10 recorders with 24-bit dynamic range and 10 recorders with 16 bit dynamic range. For all stations the sampling rate was 125 sps. Sensors were three components with 2 Hz eigenfrequency. The OBS network had an effective dynamic range of 16 bits and the sensors were 3 geophones of a 4.5 Hz eigenfrequency. The network operated from August 18 to September 20, 2000. The active phase corresponds to the period from 10 to 16 September with air gun shots every 60 s. During this period, passive seismic events had to be discriminated from the shot records. Continuous scanning of the best land records result in the creation of a catalogue which was used to extract the records on the other stations. We detected 300 passive events, however only 181 events could be located and only five events were found in the RENSIG catalogue, including one event of magnitude 4.5, also detected by the USGS network.

4. Locations

The high quality of the shot data allowed building well-constrained velocity models. Details of this work were given in previous works (Graindorge et al., 2004; Gailler, 2005; Gailler et al., 2007). One structural characteristic of this area is the thick oceanic crust (Sallares and Charvis, 2003) due to the presence of the Carnegie Ridge; other results are the oceanic crust-type velocity of the upper plate (6.1–6.4 km/s), due to the accretion of several oceanic blocks to the Andean continental margin, and finally the presence

of a low-velocity zone in the subduction channel (Graindorge et al., 2004). We also deduced the presence of the western limit of the sedimentary Manabi basin and a slab dipping about 10° between 4 and 15 km depth. We built a 3D grid based on the projection of the 2D velocity models and their onshore extrapolation on a main perpendicular profile, following the methodology described in Gailler et al. (2007). In this computation, we also consider small lateral velocity variations highlighted by the profiles parallel to the margin (Gailler, 2005). We chose a grid of 5 km resolution in the horizontal and vertical directions. The eastern part of the model is extrapolated from the structural study of the Manabi basin (Daly, 1989; Deniaud, 2000).

4.1. Preliminary locations

First, we determined preliminary locations using the “Hypo-ellipse” code developed by Lahr (1999). This location technique allows the use of several velocity models corresponding to different seismogenic regions and different stations. The code derives an arrival time table, based on the introduction of velocity gradients between the different velocity zones, obtained by linear interpolation. The “shooting rays” technique allows us to solve the propagation equations for this heterogeneous medium. Tables 1a–c lists the different velocity models in correlation with distances to the trench. Fig. 4 shows the locations obtained, with a quality factor derived from location errors calculated by the code. Uncertainties range from less than 1.5 km to more than 50 km. The average RMS is 0.64 for the entire location catalogue. The location quality for the different axes of the network is good but decreases very quickly as the distance from the major axis increases. This shows the first-order effect of network geometry versus the velocity model in the location process. We therefore divided the area in two zones: Zone 1 with local events, rather precisely located and Zone 2 on the outside of the network, corresponding to poorly located events.

During the period of the experiment only five events were common both to the RENSIG catalogue and SISTEUR network. All of them belong to events of Zone 2, corresponding to poor quality locations (RMS of 0.9) due to their distance from our network. RENSIG locations are also given with uncertainties due to the poor azimuthal coverage of the network, the significant distance to the seismic stations and significant anomalies of propagation. These anomalies may result from the presence of hot material around the volcanoes, where most seismic stations are located. Data collected from both networks were used to locate these events in order to obtain better locations and to compare the results of the two networks (Table 2): even the epicenter parameters are almost similar the depth is poorly constrained due to the poor azimuthal coverage.

The 14th September event with a magnitude of 4.5°MI was also recorded by the USGS (E5 on Fig. 4). This teleseismic location differs from ours in latitude and in depth (Table 2) due to regional models used by the location method.

4.2. HypoDD relative locations

We selected the events of Zone 1 between 1°S and 2°S and between 80.5°W and 81.5°W (Fig. 4 and Fig. 5b). Hypocenters of these events are projected onto the wide-angle models. Most events are located near the interplate zone and are deeper than ~ 10 km. A seismic nest is observed in both the overriding plate and the subducting plate and located at the border of the Manabi sedimentary basin. However this nest is not well constrained in depth and induces a linear vertical distribution of hypothetical hypocenters.

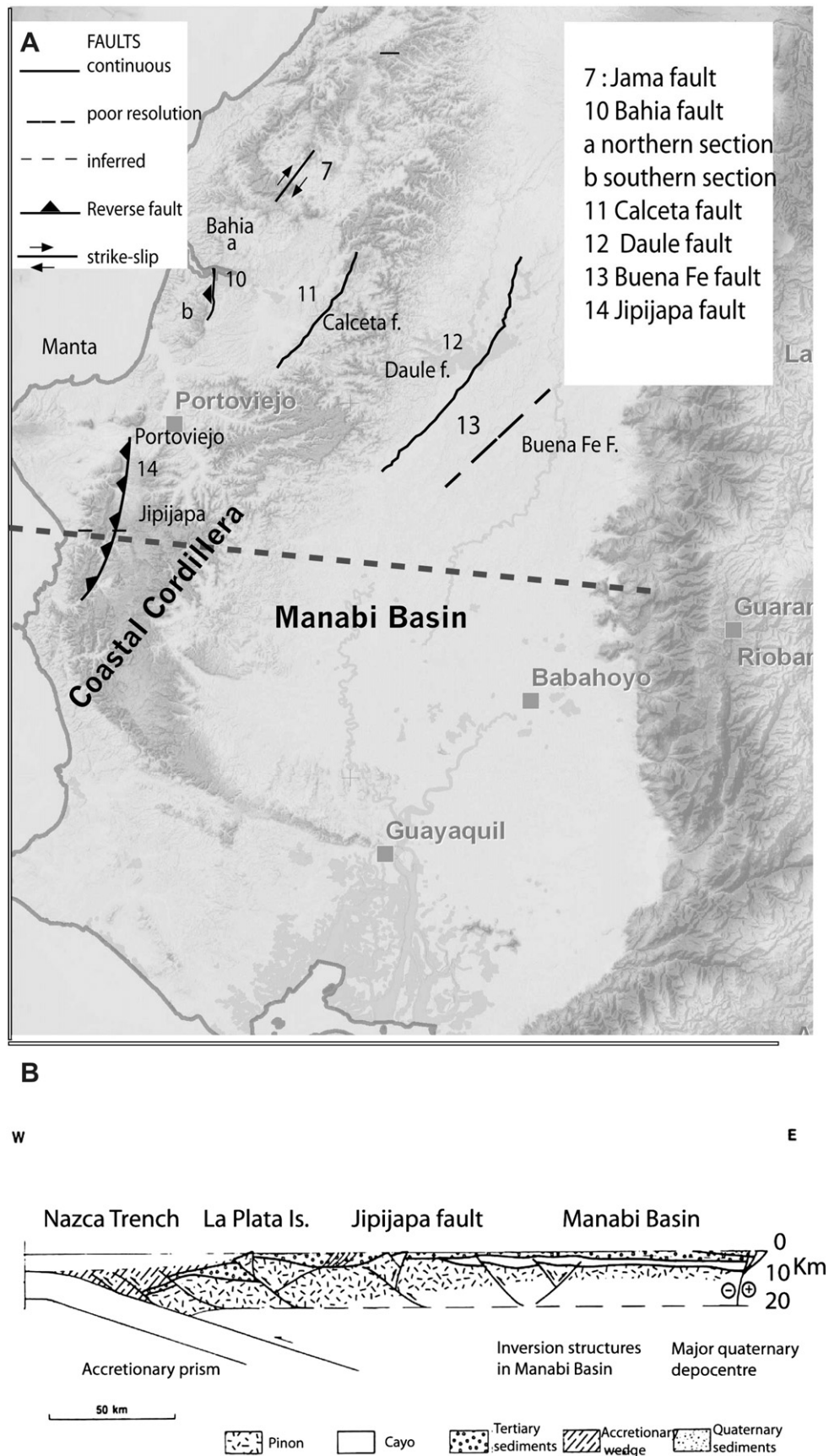


Fig. 3. A. Neotectonic map of the central North Andean block. The faults are extracted after Egúez et al., 2003. B. Structural cross-section modified from Daly (1989). The location of this profile is indicated in the map by a dotted line.

Table 1a

The three 1D velocity models are deduced from the 2D models of [Graindorge et al. \(2004\)](#) and [Gailler \(2005\)](#) obtained from shot arrival times inversion and 2D tomography. The low-velocity layers are denoted in bold. These velocity models are used for location with Hypoellipse code.

Oceanic model			Margin model			Coastal model		
Layer number	V_p (km/s)	Depth (km)	Layer number	V_p (km/s)	Depth (km)	Layer number	V_p (km/s)	Depth (km)
1	2.20	0.00	1	2.20	0.00	1	2.40	0.00
2	3.20	1.00	2	4.50	1.00	2	4.50	1.00
3	2.60	2.50	3	6.30	3.50	3	5.00	2.50
4	3.20	3.50	4	3.50	7.00	4	6.00	4.00
5	5.20	4.50	5	5.20	8.00	5	6.20	10.00
6	6.25	6.50	6	6.50	10.00	6	6.40	12.00
7	6.75	9.00	7	6.75	12.00	7	6.60	16.00
8	7.25	12.00	8	7.20	15.00	8	5.50	18.00
9	8.00	19.00	9	7.40	18.00	9	6.50	21.00
10	8.50	47.00	10	8.00	21.00	10	7.10	23.00
			11	8.50	50.00	11	8.00	30.00
						12	8.50	60.00

Bold values represent parameters of the low velocity layer.

Table 1b

1D velocity model used by RENSIG.

Layer number	Depth (km)	V_p (km/s)
1	0	3.32
2	3	5.90
3	15	6.20
4	30	6.70
5	50	8.10

Table 1c

1D velocity model used for the seismogram ($V_p/V_s = 1.71$) modeling of eastern events ($V_p/V_s = 1.74$).

Layer number	Depth (km)	V_p (km/s)
1	0	2.24
2	2	3.50
3	5	5.00
4	7	6.00
5	12	6.50
6	20	7.00
7	23	7.50
8	38	8.00

In order to improve the locations of this cluster, we use a relative location method, the so-called HypoDD method ([Waldhauser and Ellsworth, 2002](#)). This code allows the simultaneous relocation of large numbers of earthquakes, combining P and S-wave travel-time differences from catalogue data and minimizes residual differences for pairs of earthquakes by adjusting the vector difference between their hypocenters. We first used a 1D velocity model. Because the medium is varying from the trench up to the most eastern land stations, we used an average 1D model corresponding to that of the center of the study area ([Table 1a](#)). Despite this 1D approximation, the location of the swarm is strongly improved. [Fig. 5c](#) displays a more concentrated cluster and allows its division into two groups: one located in the overlying margin, the other located in the subducting plate.

4.3. Ray tracing in a 3D medium

Because we benefit from a pseudo-3D model, we look for improvement in the locations. Velocity parameters are fixed and ray paths are computed in the 3D medium using the shooting ray tracing method ([Virieux et al., 1988](#)). In this method the initial velocity model is transformed into squared slowness, which is the output parameter of the inversion. The shooting paraxial method

solves the ray equation in a smooth medium obtained by cubic interpolation of slowness. The resulting 3D medium provides a gradual variation of velocity pattern, more realistic than a velocity function varying by sharp steps. This code was already used for different studies of local earthquake tomography ([Ghose et al., 1998](#); [Haslinger and Kissling, 2001](#); [Béthoux et al., 2007](#)). Here, we chose the starting hypocenters' parameters obtained from "Hypoellipse" locations, in order to test the performances of relative hypoDD locations with respect to the location of single events in a 3D model. Hypocenter parameters are then inverted and the new hypocentral distribution obtained for the short-range seismicity is presented in [Fig. 5d](#). We superimposed the relocation of the seismic nest onto the cross-section of the synthetic velocity model. Even, if the overall variance is only slightly improved from 0.5 up to 0.4, the distribution of hypocenters shows only a very few shifts with respect to the previous results obtained with a 1D model.

5. Magnitude evaluation

Using the magnitudes of the 5 events registered by both networks ([Table 2](#)), we developed an approximate local magnitude scale for all the events, comparing the recorded amplitudes and employing the general equation:

$$M_i(i,j) = \alpha_j \log[A(i,j)] + \beta_j \log[D(i,j)] + C_j$$

where i is the seismic event at the j the station, A is the amplitude, D is the epicentral distance between i and j , α_j and β_j are empirical coefficients and C_j is a term which depends upon the station. A magnitude scale was derived by comparing the recorded amplitudes for each station. During the short operation period we recorded preliminary magnitudes ranging from 2.0 up to 4.5, being the reference, the biggest event of magnitude of 4.5 located both by the USGS and RENSIG network.

6. Focal mechanisms and modeling

6.1. Focal mechanisms

After verifying the polarities of our sensors by means of air gun recording, we determined preliminary focal mechanisms, using polarities from our data and those of RENSIG when available. Focal solutions were obtained employing the FPFIT code ([Reasenberger and Oppenheimer, 1985](#)), which computes all solutions compatible with the distribution of polarities and gives the corresponding strike and dip uncertainties.

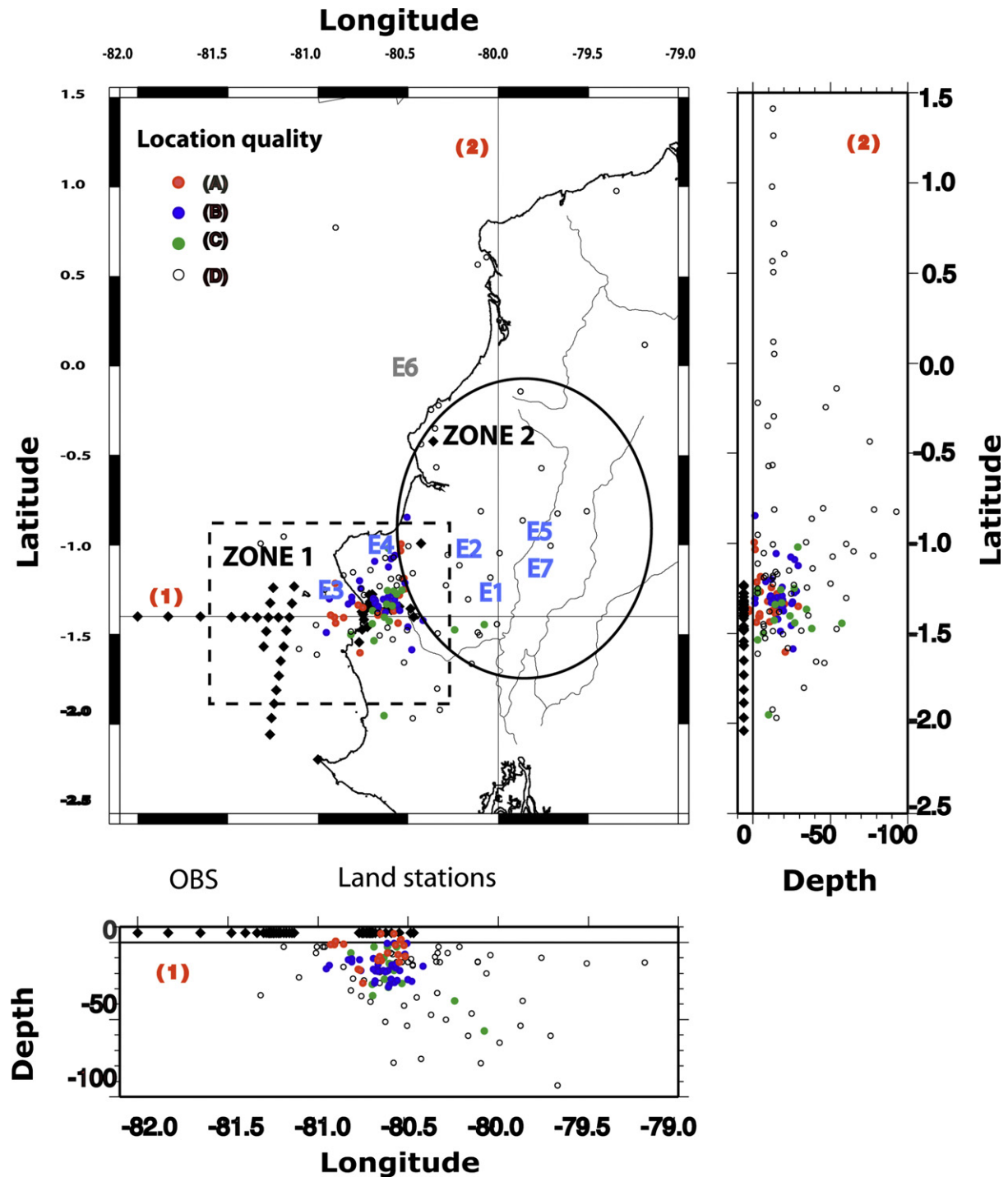


Fig. 4. Preliminary location of events recorded during SISTEUR experiment. The quality location (Max SEH, SEZ) is indicated by colors [A-red: ≤ 1.34 km, B-blue: ≤ 2.64 km, C-green: ≤ 5.35 km, D-white: > 5.35 km]. The studied region has been divided into two zones: Zone 1 near the network with well located events, Zone 2 corresponds to farther ill-located events. The events denoted E are those studied by waveform modeling.

For seismicity of Zone 1, recorded with only the SISTEUR network, the focal mechanisms are poorly constrained. However, the deeper located events correspond to reverse solutions with an E-trending P-axis. For the three events (R1, R3, R4) well located in the interplate zone, we chose a common focal solution, compatible with the three different distributions of polarities, that is a pure reverse faulting solution with one nodal plane characterized by an eastward dip of $\sim 30^\circ$. For the two shallower events (R2 and R5) we

obtained identical solutions, a transpressional solution, with an E-trending P-axis and an N-trending T-axis, whose nodal plane is similar to the strike of the Jipijapa fault. These solutions are reported in Fig. 7 and Table 3. For more distant events (Zone 2) that were also recorded by the stations of the RENSIG network, we propose better focal solutions, which were computed with the FPFIT code as starting parameters for the modeling described in the next section.

Table 2
Comparison of location for events detected by the RENSIG network. The comparison shows a rather good agreement in epicenter coordinates but a big discrepancy in depth.

NETWORK	DATE	H0	Latitude (°S)	Longitude (°W)	Depth (km)	R.M.S.	Mag.
SISTEUR	00/08/27	00:54:45	1.224	80.2898	47.20	0.118	
RENSIG			1.203	79.895	16.00	0.765	4.1
SIS + REN			1.205	80.020	8.54	0.288	
SISTEUR	00/08/28	19:48:30	0.823	79.672	89.90	1.840	
RENSIG			0.830	79.812	28.06	0.785	4.1
SIS + REN			0.900	79.726	138.00	0.293	
SISTEUR	00/09/01	05:56:49	0.568	79.760	7.20	0.799	
RENSIG			0.996	79.289	118.80	0.472	3.4
SIS + REN			0.520	79.760	76.81	0.547	
SISTEUR	00/09/14	12:46:22	0.863	79.865	35.00	0.303	
RENSIG			0.793	79.734	32.85	0.420	4.5
USGS			0.550	79.700	33.00		4.5
SIS + REN			0.927	79.725	8.00	0.784	
SISTEUR	00/09/16	03:24:26	1.111	80.216	1.50	0.511	
RENSIG			1.436	79.650	24.32	0.542	3.8
SIS + REN			1.152	79.726	30.37	0.607	

6.2. Regional range seismicity modeling

We studied the waveform of available records in order to better constrain the location of some events, located rather far for the network. Indeed, for regional distances the waveform is mainly related to the hypocentral parameters, and in a secondly to the focal mechanism (Bertil et al., 1989). We calculated synthetic waveforms using the discrete wave-number method implemented by Bouchon and Aki (1977) and the code modified by Coutant (1994) who replaced the computation of wave propagation at the interface obtained with Thompson–Haskell methodology, by a matrix computation of reflection and transmission coefficients at each interface of a one-dimensional velocity model. The so-called AXITRA code computes the Green solutions in the frequency domain and depends upon the hypocentral coordinates the position of the station with respect to the hypocenter, and the crustal model (velocity, density, Q factors and thickness of each layer). These Green solutions are then convoluted with the source function, the focal mechanism and seismic moment M_0 . Afterwards, the calculated seismograms are compared with the observed records in the time domain.

In our case the model is strongly 2D in a west-east direction. So, in order to validate the condition of $\sim 1D$ model between the source and the receivers, we first limit the computation to the ray path between the source and the stations, which are approximately parallel to the continental margin. Events E3, E4 and E6 whose location is reported in Fig. 8 obey this condition. The crustal model used for this modeling depends upon the position of the ray path respect to the margin. The other studied events (E1, E2, E5 and E7) are located outside the studied wide-angle profile, at the border of the Manabi sedimentary basin (Fig. 8). Consequently, we extrapolated the crustal model of Fig. 5, taking into account the presence of volcanoclastic deposits of the Cayo formation (Daly, 1989) in the basin and the eastward thickening of the overlying margin. Table 1c shows the resulting velocity model.

The Green functions computed in the frequency range [0.5–10 Hz] are then convoluted by a source function obtained from the postulated focal mechanisms, a seismic moment M_0 inferred from the evaluation of M_l and a temporal rise of the time function compatible with the magnitude of the event.

First we checked the quality of epicentral parameters by verifying that the synthetic seismogram, obtained in the time domain, provides S–P arrival times compatible with the one observed. Then we tried to better constrain the depth range. Because the duration of the L_g phase and more generally the shape of the P and S envelopes are both strongly linked to the depth, we checked the

different depth values for the Green function computation (by steps of 20 km from 0 up to 80 km, then by steps of 5 km around the approximate depth). We then fixed the Green functions for a chosen depth. Finally, we analyzed the focal mechanism because it has a strong influence on the S amplitude respect to P amplitude ratio and on the first P arrival waveform. This synthetic seismogram is obtained by convolution of the chosen Green function with different source parameters, verifying the starting focal solution obtained previously with the distribution of P-wave polarities. Because we used only short-period sensors and rather noisy records, only the general waveform could be studied here. We focused on the coda shape (mainly related to the depth focus) and on the P/S ratios (mainly related to the focal mechanism). The best solutions correspond to the best cross-correlations between the observed and the synthetic seismograms. Fig. 6 depicts some comparisons between observed and synthetic seismograms. Table 3 shows the constrained hypocenters and focal solutions obtained from these waveform models.

7. Interpretation

Despite the short recording period, several interesting results can be deduced from this study. The seismicity is located both in the overriding plate and in the subducting one. We note that a rather shallow seismicity is present in this region, whereas no events were recorded in the overriding plate north of the Esmeraldas region during the “SUBLIME” experiment, using the same methodology (Pontoise and Monfret, 2004).

Some events are located just on the intraplate boundary, as deduced from the wide-angle modeling. The minimum depth of these events is ~ 10 km, which likely characterize the upper limit of the ISZ along the southern flank of the Carnegie Ridge (Fig. 5). Farther east the hypocenters get deeper, up to 35 km at a distance of ~ 100 km from the trench. The events located in the subducting plate imply reverse faulting (events R1, R3, R4, E3) with an E-trending P-axis and a nodal plane dipping $\sim 30^\circ$ (Figs. 7 and 8), whereas Graindorge et al. (2004) determined a subducting slab dipping 10° east between 4 and 15 km depth, from their profile modeling. Other studies also found an average slab dip of about 30° (Guillier et al., 2001; Pontoise and Monfret, 2004). We therefore infer that the events R1, R3, R4, located deeper than 15 km, can correspond to events located at a curve of the slab, farther from the trench. Part of this discrepancy can also be due to uncertainties in focal parameters. Event E3, located nearer the trench may correspond to a local rupture of a seamount as discussed by Vaca et al. (2010) for the 2005 Manta crisis. The source study of the main

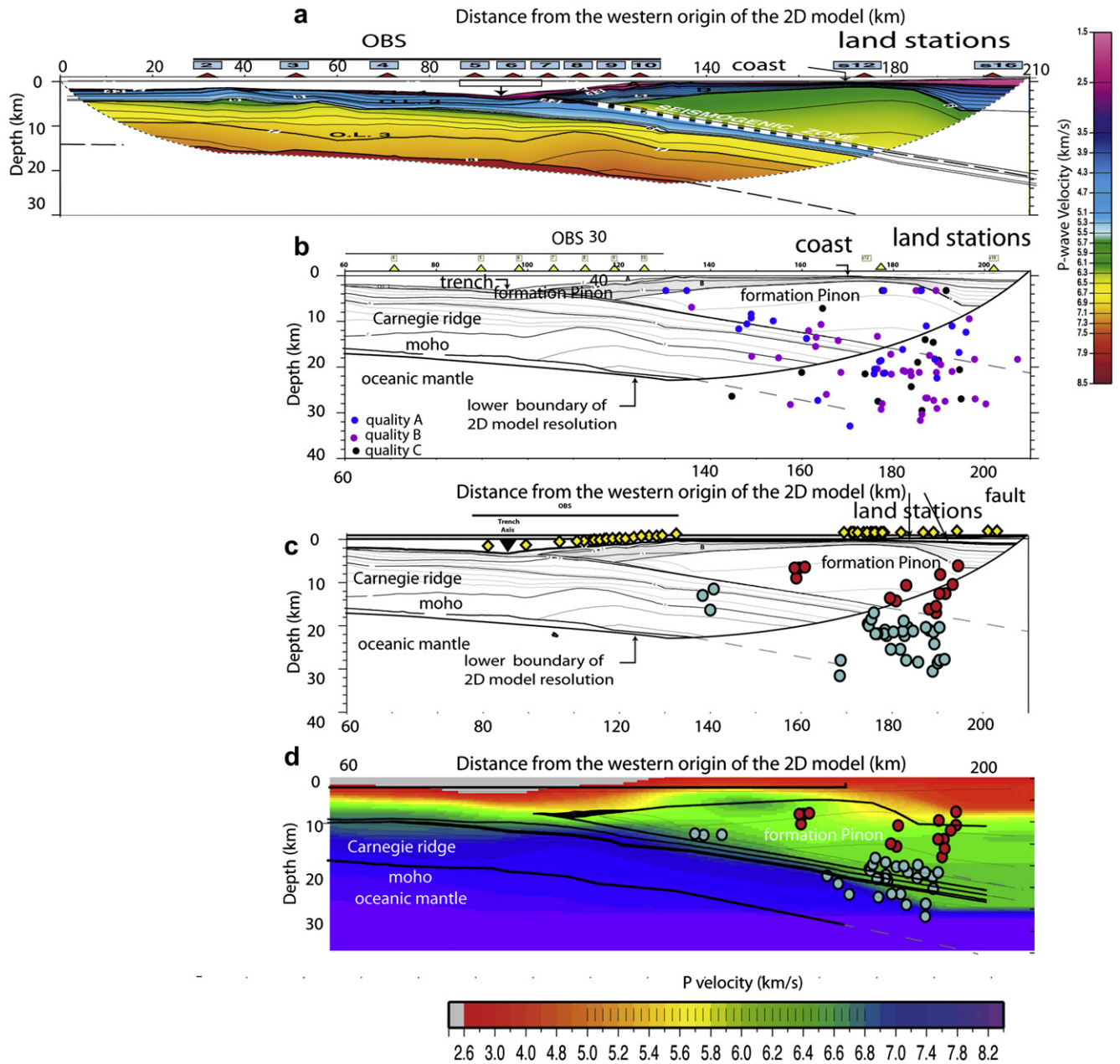


Fig. 5. a. 2D velocity model obtained from the shots data of Graindorge et al. (2004). b. Events of Zone 1 are projected onto the cross-section corresponding to this velocity model. c. Relocation of some events using the HypoDD method (Waldhauser and Ellsworth, 2002). d. Relocation of the events with the interpolated 3D model.

Table 3

Revised hypocentral locations and focal solutions obtained thanks to seismogram modeling.

N°	Date	T ₀ [HH:Mn]	LAT [°S]	LON [°W]	Z	Az1 [°N]	δ1 [°]	λ1 [°]	Az P [°N]	ΔP [°]	Az T [°N]	ΔT [°]	MI
R1	00/09/02	11:06	1.464	80.639	19	10	30	90	280	15	100	75	3.1
R2	00/09/05	18:41	1.361	80.711	20	10	30	90	280	15	100	75	2.8
R3	00/09/11	12:42	1.033	80.524	5	200	50	145	74	8	174	50	2.9
R4	00/09/11	19:36	1.213	80.540	19	10	30	90	280	15	100	75	3.5
R5	00/09/13	13:59	1.221	80.510	7	200	50	145	74	8	174	50	3.0
E1	00/08/27	00:54:45	1.210	80.020	10	230	50	-166	82	36	186	19	4.1
E2	00/09/04	00:55:13	1.010	80.200	5	105	60	-83	33	74	190	15	3.2
E3	00/09/10	03:22:42	1.230	80.940	10	350	29	26	306	28	176	50	2.9
E4	00/09/13	12:10:18	0.960	80.610	8	200	50	145	74	8	174	50	3.5
E5	00/09/14	12:46:22	0.830	79.720	5	310	70	-47	264	46	10	14	4.5
E6	00/09/14	17:15:01	0.020	80.560	15	22	20	90	292	25	112	65	3.4
E7	00/09/16	03:24:20	1.150	79.730	17	208	65	125	274	13	164	55	3.8

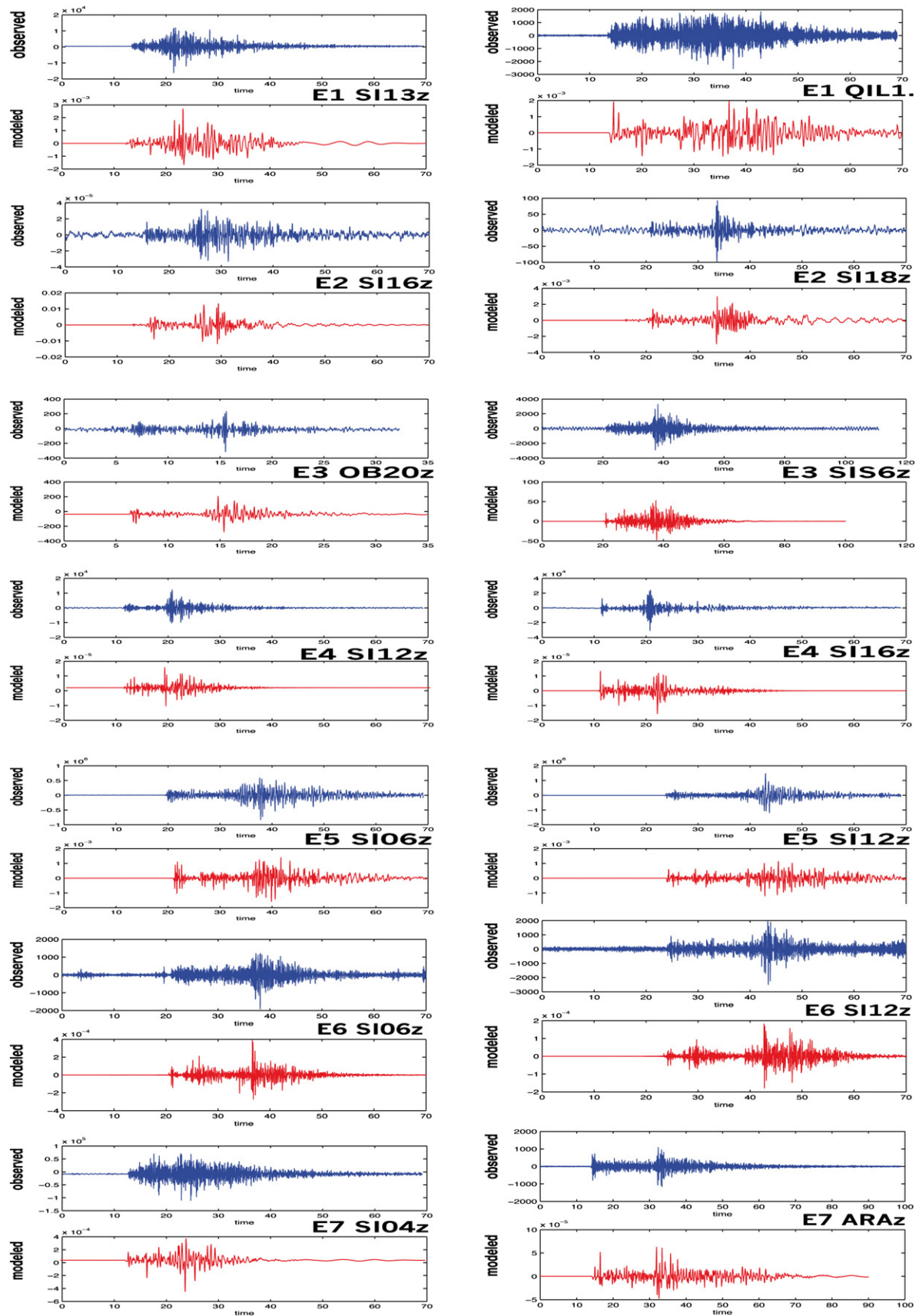


Fig. 6. Examples of waveform modeling for events denoted E in Fig. 4. The observed seismogram is above the modeled one. The source parameters are displayed in Table 3. Stations denoted SI are stations of the SISTEUR network and OB20 is the OBS number 20 (see Fig. 2), QIL1 (0° 49.32'S, 78° 27.31'W) and ARAZ (1° 30.38'S, 78° 56.08'W) are RENSIG stations also modeled.

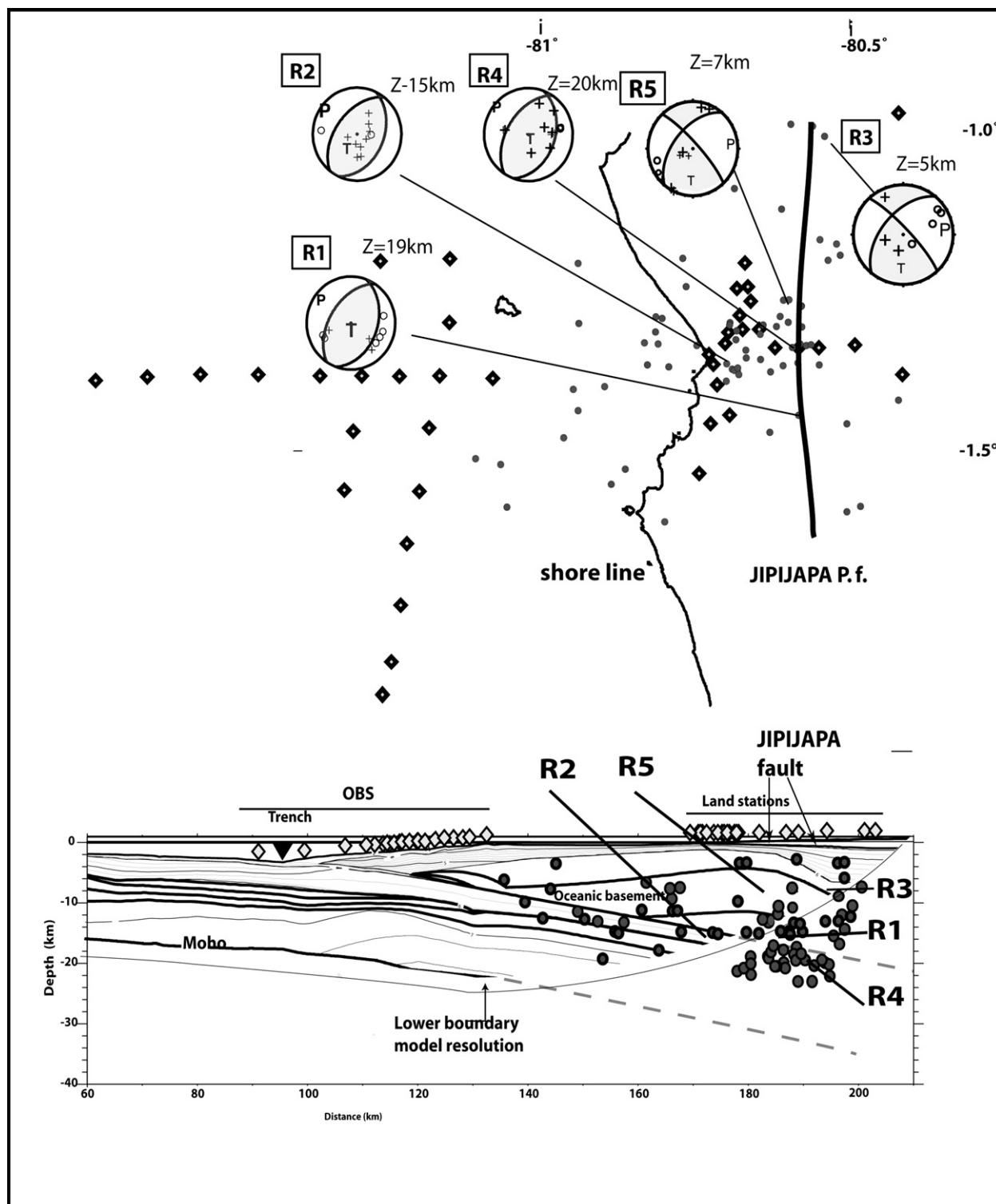


Fig. 7. Location and focal mechanisms of events of Zone 1. The stations are indicated by diamonds, the epicenters by circles. The numbers correspond to Table 3.

quakes of the Manta crisis, using the method of Nabelek (1984) allows us to confirm this range of focal depths (with an updip limit of ISZ around 10 km) and nodal planes ranging from 10° up to 24° (Vaca et al., 2010). The event E6, North of the Carnegie Ridge, is located near the epicenter of the 1942, 7.9 Ms event, North of Bahía de Caráquez. Mendoza and Dewey (1984) relocated this event at a depth of 19.7 km and the shock of 1958 ($M_s = 7.3$) at a depth of

21.7 km (Figs. 1 and 8). According to the waveform model the E6 event has an inferred depth of ~ 15 km (± 2 km) and a nodal plane characterized by an eastward dip of $\sim 20^\circ$ and P-axis trending E–W, in good agreement with the solutions given by Kanamori and MacNally (1982) for both the 1942 and 1958 events. These focal solutions are compatible to the general compressive regional state along the interplate zone in Colombia–Ecuador, with a σ_1 trending

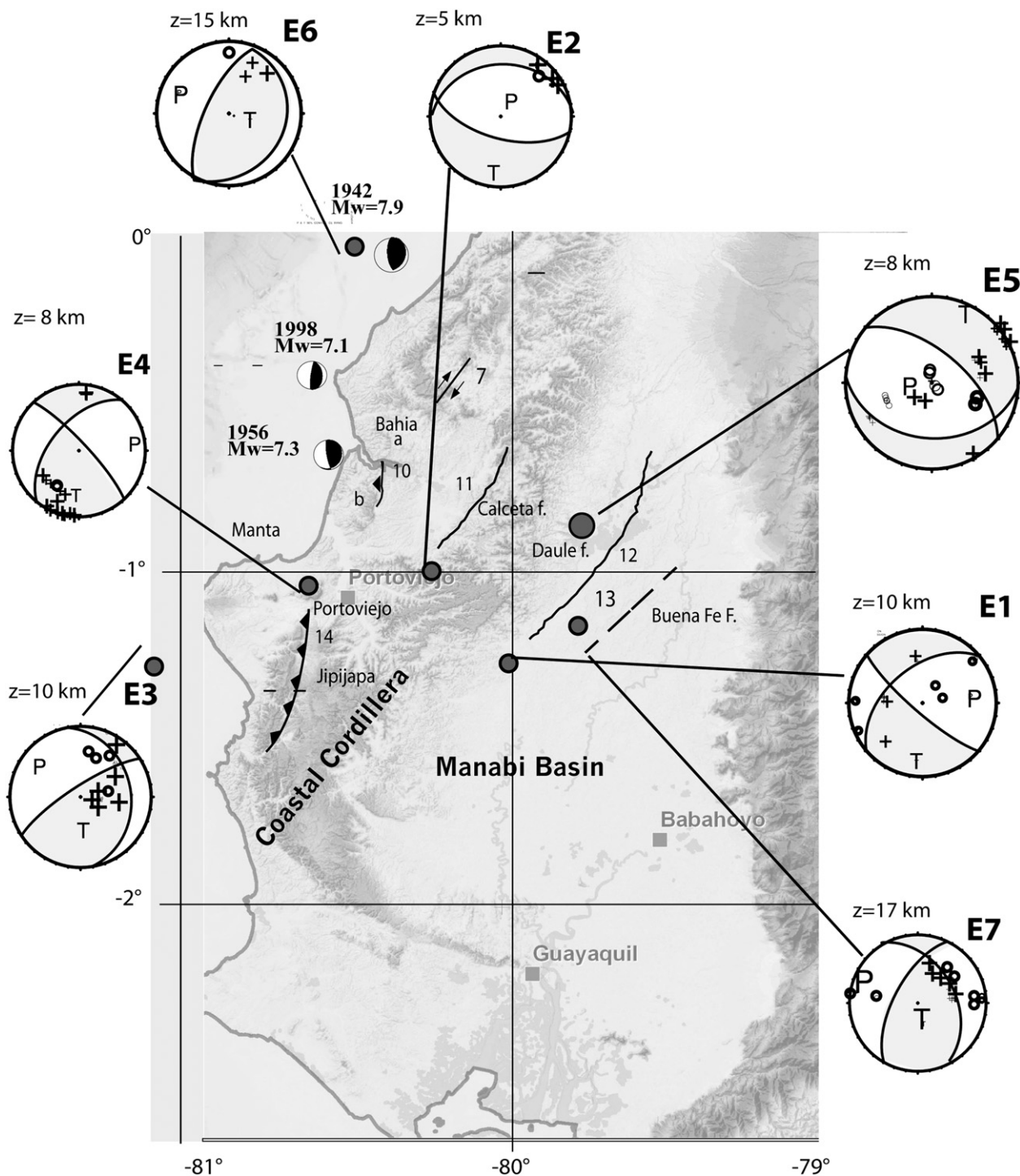


Fig. 8. The relocated events of Zone 2 are reported in the Ecuadorian Neotectonic Map Ecuador. For the faults cited in the text, the numbers correspond to the description of the Quaternary faults and/or lineaments in Eguez et al. (2003). Seven focal mechanisms could be established (solutions are in Table 3). Location of the 1942 earthquake is from Beck and Ruff (1984), location of the 1958 earthquake is from Mendoza and Dewey (1984) and the 1998 earthquake is from Harvard catalogue.

from N80° (Ego et al., 1996) to N121.1° according to Corredor (2003). Legrand et al. (2005) computed a stress tensor more typical of the central coastal zone of the Ecuadorian margin, inverting focal mechanisms available in the CMT Harvard catalogue. The deduced σ_1 has a strike of N265° and a dip of 9°. The P axes deduced from our mechanisms are close to this orientation.

However the main result is the presence of a two nests of seismic events, one in the overriding plate, the second in the

subsiding crust, near the seismic network. This cluster location corresponds to the trace of the Portoviejo-Jipijapa fault already described in Section 2 and Fig. 3. This study shows that this fault is seismically active and possibly deeply rooted in the crust reaching the interplate boundary. The event E4, located north of the network, was studied in greater detail; its focal depth of 7 km determined by location was first tested. We verified that this value permit us to compute synthetic waveforms similar to the ones

observed in several stations. Because insufficient readable P-wave polarities are available, (see Fig. 7) we tested several solutions from pure reverse to normal focal mechanisms. A solution showing a nodal plane compatible with a dextral fault oriented N20° describes well the observed waveforms. This solution is the same as that chosen for events R2 and R5 that is a transpressive movement as inferred for the Jipijapa fault in the Ecuadorian Neotectonic Map (Egüez et al., 2003) noted as 14 in this map (Fig. 8).

East of the network, we mainly located events in the overriding plate. Event E7 corresponds to a well-constrained inverse mechanism with an NNE–SSE-trending fault plane, compatible with the orientation of Quaternary faulting as given in the Ecuadorian Neotectonic Map, and especially with the Buena Fe fault, which forms a weak lineament on radar images (No. 13 in Fig. 3). Modeling is in agreement with this mechanism. The source parameters of event E1 are less well-constrained, but the distribution of P-wave polarities as well as the modeling allows us to infer a normal component dextral strike-slip faulting trending in an N30° direction, in good agreement with the orientation of the Daule fault (No. 12 in Fig. 3); its epicenter is very close to the superficial expression of this fault already associated with some seismicity (Egüez et al., 2003).

The largest magnitude event (E5) recorded during the experiment is located on the border of the Manabi basin. Computing synthetics allowed us to validate the new epicenter coordinates (SIS + REN in Table 2) and to constrain its depth checking values from 33 km (RENSIG and USGS catalogues) up to 2 km. A depth of 5 km provides realistic synthetic waveforms at different azimuths. For this event and for the nearby E2 event we found normal solutions corresponding to rupture planes orientated NW–SE dipping to the south. While the location of E5 event is also close to the Daule fault, it is more likely associated with the activity of a blind fault hidden under the Manabi basin, possibly associated with the Bahía de Caráquez fault (Fig. 3), as postulated by Daly (1989) from geodynamic reconstitution. The E2 event might also correspond to another parallel fault hidden under the Manabi basin.

8. Tectonic and structural conclusion

The lack of a well-distributed and permanent seismic net and the corresponding poor knowledge of local seismicity led us to develop a methodology based on the knowledge of the velocity of the Ecuadorian margin derived from SISTEUR experiment. Despite its significant seismic activity expressed by several important seismic crises including the 2005 events (Vaca et al., 2010), the seismic activity and tectonics of the Manta-Puerto López region are not yet well known. The present preliminary study, covering only the background seismicity, is a contribution towards a better understanding of the active central Ecuadorian margin.

The upper limit of the Interplate Seismogenic Zone (ISZ) in this part of the margin is around 10 km. The seismic activity of the overriding plate provides new insights:

- 1) Seismic activity that reaches the base of the overriding plate is located along the trace of the Jipijapa-Portoviejo fault, defined as a dextral strike-slip fault with a reverse component. Several focal solutions computed for events of this next exhibit rupture planes parallel to its superficial projection (N10°–N25°). These results are in agreement with the present-day activity of this fault;
- 2) Two studied events are located east of the Coastal Cordillera and have focal mechanisms with nodal planes, approximately parallel to quaternary structures reported in the Ecuadorian Neotectonic Map (Egüez et al., 2003). Two other focal mechanisms exhibit normal faulting and a nodal plane of N110° dipping to the south. For the lack of known lineaments in this area, we suggest that

this activity belongs to blind faults, hidden under the Manabi basin, possibly inherited from Oligocene orientations indicated by Daly (1989);

- 3) P and T axes computed for the 13 focal mechanisms presented in this study are in agreement with solutions that are coherent with a homogeneous stress field produced by the convergence of the Nazca and South American plates and the escape of the North Andean Block towards the NNE. Because this movement is faster in the north (7.0–10.0 mm/yr) than in the south of the NAB (3.6 mm/yr), according to neotectonic and geodetic results (Winter et al., 1993; Ego et al., 1996; Legrand et al., 2005; Tibaldi et al., 2007; Nocquet et al., 2010), this discrepancy could involve N–S to NW–SE extension (E1, E2, E5 solutions) have a normal component. These normal solutions are compatible with the strike-slip solutions, which seem to characterize the movements of faults orientated NE–SW, such as the Jipijapa fault and other faults located near the border of the Coastal Cordillera. Normal component mechanisms along faults orientated WNW–ESE might be part of a pull-apart system between parallel dextral strike-slip faults that might be expected by the movement of the NAB towards the NE. So far this normal deformation has been shown only in the coast (Pedoja et al., 2006) or farther south in the Guayaquil region (Dumont et al., 2005). From their study of industrial wells and seismic profiles, Witt et al. (2006) show that the tectonics of the Guayaquil region is dominantly extensional at the present time. They conclude that the N–S tensional stress regime of the Gulf of Guayaquil should be the effect of north-eastward drift of the NAB. The presence of normal fault mechanisms, in the region north of the Gulf of Guayaquil should document this hypothesis. Consequently, the North Andean Block should not be considered as a rigid one-piece plate escaping to the NNE, without internal deformation, at least not in the transitional part of the two regions of the margin previously described i.e. the northern region with major subduction events and the southern region with only moderate seismicity).

This study indicates the urgent need for more seismic stations to monitor both the interplate margin and the North Andean Block between the coast and the Andes Cordillera. By studying the background seismicity over a longer period of time should clarify the response of the upper plate to the subduction of the Carnegie ridge. The deformation of the NAB would greatly benefit from GPS monitoring and the resulting better geodetic models.

Acknowledgements

This paper is published in memory of Bernard Pontoise who passed away before this study was finished. The SISTEUR campaign was supported by Institut de Recherche pour le Développement (IRD), Institut National des Sciences de l'Univers (INSU-CNRS) and Institut Français pour l'Exploitation de la Mer (IFREMER) which provided ship time. Also special thanks to Francis Boudoux and Piedad Martinez for their precious help in the field and to Yann Hello, Alain Anglade and Ben Yates who operated the OBS network. We are grateful to the two anonymous reviewers and the editor who helped us in improving the manuscript.

References

- Beck, S.L., Ruff, L.J., 1984. The rupture process of the great 1979 Columbia earthquake: evidence for the asperity model. *Journal of Geophysical Research* 89, 9281–9291.
- Benítez, S., 1995. Evolution géodynamique de la province côtière sud-équatorienne au Crétacé supérieur-Tertiaire. Université Joseph Fourier (Grenoble I), Grenoble, France, pp. 208.

- Bertil, D., Bethoux, N., Campillo, M., Massinon, B., 1989. Modeling crustal phases in southeast France for focal depth determination. *Earth and Planetary Science Letters* 95, 341–358.
- Béthoux, N., Sue, C., Paul, A., Virieux, J., Cattaneo, M., Fréchet, J., Thouvenot, F., 2007. Local tomography and focal mechanisms in southwestern Alps: comparison of methods and tectonic implications. *Tectonophysics* 432, 1–19.
- Bouchon, M., Aki, K., 1977. Discrete wave-number representation of seismic-source wave fields. *Bulletin of the Seismological Society of America* 67 (2), 259–277.
- Collot, J.-Y., Marcaillou, B., Sage, F., Michaud, F., Agudelo, W., Charvis, P., Graindorge, D., Gutscher, M.A., Spence, G., 2004. Are rupture zones limits of great earthquakes controlled by upper plate structures? Evidence from multi-channel seismic reflection data acquired across the northern Ecuador–Southwest Colombia margin. *Journal of Geophysical Research* 109, B11103. doi:10.1029/2004JB003060.
- Collot, Y., Charvis, P., Gutscher, M.-A., Operto, S., 2002. Exploring the Ecuador–Colombia active margin and interplate seismogenic zone. *EOS Transactions, American Geophysical Union* 83 (no 17) 185, 189, 190.
- Corredor, F., 2003. Seismic strain rates, distributed continental deformation, and three-dimensional seismotectonics of northwestern South America. *Tectonophysics* 372, 147–166.
- Coutant, O., 1994. AXITRA Program: Expression of the Green's Functions in Cylindrical Coordinate Used with a Reflectivity Method. LGIT IRIGM, Grenoble, France.
- Daly, M.C., 1989. Correlations between Nazca/Farallon plate kinematics and forearc basin evolution in Ecuador. *Tectonics* 8 (4), 769–790.
- Deniaud, Y., 2000. Enregistrement sédimentaire et structural de l'évolution géodynamique des Andes équatoriennes au cours du Néogène: étude des bassins d'avant-arc et bilans de masse. Université Joseph Fourier (Grenoble I), Grenoble, France, pp. 242, PHD.
- Dumont, J.F., Santana, E., Vilema, W., Pedoja, K., Ordóñez, M., Cruz, M., Jiménez, N., Zambrano, I., 2005. Morphological and microtectonic analysis of Quaternary deformation from Puna and Santa Clara Islands, Gulf of Guayaquil, Ecuador (South America). *Tectonophysics* 339, 331–350.
- Ego, F., Sebrier, M., Lavenue, A., Yepes, H., Egüez, A., 1996. Quaternary state of stress in the northern Andes and the restraining bend model for the Ecuadorian Andes. *Tectonophysics* 259, 101–116.
- Egüez, A., Alvarado, A., Yepes, H., 2003. Mapa de Fallas Activas y Pliegues Cuaternarios de Ecuador y Regiones Oceánicas adyacentes. <http://pubs.usgs.gov/of/2003/ofr-03-289/>.
- Gailler, A., 2005. Structure de la marge d'Equateur-Colombie par modélisation des données de sismique grand-angle marines. Université de Nice-Sophia Antipolis, Nice, France, 257 p, PHD.
- Gailler, A., Charvis, P., Fluh, E., 2007. Segmentation of the Nazca and South American plates along the Ecuador subduction zone from wide angle seismic profiles. *Earth and Planetary Science Letters* 260, 444–464.
- Ghose, S., Hamburger, M., Virieux, J., 1998. Three dimensional velocity structure and earthquake locations beneath the northern Tien Shan of Kyrgyzstan, central Asia. *Journal of Geophysical Research* B2 (103), 2725–2748.
- Graindorge, D., Collahorano, A., Charvis, Ph., Collot, J.Y., Bethoux, N., 2004. Deep structure of the Ecuador convergent margin and the Carnegie Ridge, possible consequence on great earthquakes recurrence interval. *Geophysical Research Letters* 31, L04603.
- Guillier, B., et al., 2001. Seismological evidence on the geometry of the orogenic system in central northern Ecuador (South America). *Geophysical Research Letters* 28 (19), 3749–3752.
- Handshumacher, D.W., 1976. Post-Eocene plate tectonics of eastern Pacific. In: Sutton, G.H., Manghnan, M.H., Moberly, R. (Eds.), "The Geophysics of Pacific Ocean Basin and Its Margin", (Wollard Volume), vol. 19. Am. Geophys. Un. Geophys. Monograph., pp. 177–202.
- Haslinger, F., Kissling, E., 2001. Investigating effects of 3-D ray tracing methods in local earthquake tomography. *Physics of the Earth and Planetary Interiors* 123 (2–4), 103–114.
- Herron, E.M., 1972. Sea floor spreading and the Cenozoic history of the east-central Pacific. *Geological Society of America Bulletin* 83, 1671–1692.
- Hey, R., 1977. Tectonic evolution of the Cocos–Nazca spreading center. *Geological Society of America Bulletin* 88 (No. 10), 1404–1420.
- Jaillard, E., Benitez, S., Mascle, G.H., 1997. Les déformations paléogènes de la zone d'avant-arc sud-équatorienne en relation avec l'évolution géodynamique, vol. 168. *Bull. Soc. Géol. France* (4): pp. 403–412.
- Kanamori, H., MacNally, K.C., 1982. Variable rupture mode of the subduction zone along the Ecuador–Colombia coast. *Bulletin of the Seismological Society of America* 72 (4), 1241–1253.
- Kellogg, J.N., Bonini, W.E., 1982. Subduction of the Caribbean plate and basement uplifts in the overriding South-American plate. *Tectonics* 1, 251–276.
- Lahr, J.C., 1999. Hypoellipse: A Computer Program for Determining Local Earthquakes Hypocentral Parameters, Magnitude, and First-Motion Pattern. USGS–Denver Federal Center, Denver, USA.
- Legrand, D., Baby, P., Bondoux, F., Dorbath, C., Bès de Berc, S., Rivadeneira, M., 2005. The 1999–2000 seismic experiment of Macas swarm (Ecuador) in relation with rift inversion in Subandean foothills. *Tectonophysics* 395, 67–80.
- Longsdale, P., Klitford, K.D., 1978. Structure and tectonic history of the esatern Panama Basin. *Geological Society of America Bulletin* 89, 981–999.
- Mammerickx, J., Herron, E., Dorman, L., 1980. Evidence for two fossil spreading ridges in the southeast Pacific. *Geological Society of America Bulletin* 1 (91), 263–271.
- Mann, P., Burke, K., 1984. Neotectonics of the Caribbean. *Reviews of Geophysics and Space Physics* 22, 309–362.
- Mann, P., Corrigan, J., 1990. Model of late deformation in Panama. *Geology* 18, 558–562.
- Mendoza, C., Dewey, J.W., 1984. Seismicity associated with the great Columbia–Ecuador earthquakes of 1942, 1958 and 1979: implications for barrier models of earthquake rupture. *Bulletin of the Seismological Society of America* 74, 577–593.
- Minster, J.B., Jordan, T.H., 1978. Present-day plate motions. *Journal of Geophysical Research* 83, 5331–5354.
- Nabelek, J., 1984. Determination of source parameters from inversion of body waves. Mass. Inst. of Technol., Cambridge, PhD. thesis.
- Nocquet, J.M., Mothes, P., Alvarado, A., 2010. Geodesia, geodinámica y ciclo sísmico en Ecuador in *Geología y Geofísica Marina y Terrestre del Ecuador. Spec. Pub. INOCAR-IRD*, pp. 83–95.
- Pedoja, K., Dumont, J.F., Lamth, M.L.O., Ortlieb, L., Collot, J.Y., Ghaleb, B., Auclair, M., Alvarez, V., Labrousse, B., 2006. Pliocaternary uplift of the Manta peninsula and La Plata Island and the subduction of the Carnegie Ridge, central coast of Ecuador. *Journal of South America Earth Sciences* 22, 1–21.
- Pennington, W.D., 1981. Subduction of the eastern Panama Basin and the seismotectonics of northwestern South America. *Journal of Geophysical Research* 86, 10753–10770.
- Pontoise, B., Monfret, T., 2004. Shallow seismogenic zone detected from an offshore–onshore temporary seismic network in the Esmeraldas area (northern Ecuador). *Geochimistry, Geophysics, Geosystems* 5, 22.
- Reasenber, P., Oppenheimer, D., 1985. FPFIT, FPLOT and FPPAGE: Fortran computer programs for calculating and displaying earthquake fault-plane solutions. *United States Geological Survey*, 85–739. Open-File Rep.
- Reyes, P., Dauteuil, O., Michaud, F., May 2010. Relative Quantification of Coastal Cordillera (Ecuador) Uplift: Preliminary Results from Quantitative Geomorphology. abstract. EGU, Vienna.
- Reyes, P., 2008. Quantification relative du soulèvement de la Cordillère cotière (Equateur) à partir de la géomorphologie sur MNT. Mem. Master2, DSGA. univ. Nice, pp. 55.
- Sage, F., Collot, J.Y., Ranero, C.R., 2006. Interplate patchiness and subduction–erosion mechanisms. Evidence from depth-migrated seismic images at the central Ecuador convergent margin. *Geology* 34 (12), 997–1000.
- Sallares, V., Charvis, P., 2003. Crustal thickness constraints on the geodynamic evolution of the Galapagos volcanic province. *Earth and Planetary Science Letters* 214, 545–559.
- Sallares, V., Charvis, P., Flueh, E.R., Bialas, J., 2005. Seismic structure of the Carnegie Ridge and the nature of the Galapagos hotspot. *Geophysical Journal International* 161, 763–788.
- Segovia, M., Alvarado, A., 2010. Breve análisis de la sismicidad en el Ecuador in *Geología y Geofísica Marina y Terrestre del Ecuador. Spec. Pub. INOCAR-IRD*, pp. 131–150.
- Segovia, M., 2001. El sismo de Bahía del 4 de Agosto de 1998: Caracterización del mecanismo de ruptura y análisis de la sismicidad en la zona costera, Tesis Ing. Geol., Escuela Politécnica Nacional, Quito, Ecuador.
- Soulas, J.P., Eguez, A., Yepes, H., Perez, H., 1991. Tectonica Activa y riesgo sísmico en los Andes Ecuatorianos y en extemo sur de Colombia. *Boletín Geologica Ecuatoriano* 11, 3–9.
- The US National Earthquake Information Center (NEIC). National Earthquake Information Center (USGS). <http://www.neic.usgs.gov>.
- Tibaldi, A., Rovida, A., Corazzato, C., 2007. Late Quaternary kinematics, slip rate and segmentation of a major Cordillera-parallel transcurrent fault; the Cayambe–Afiladores–Sibundoy system, NW South America. *Journal of Structural Geology* 29 (4), 664–680.
- Toussant, J.F., Restrepo, J.J., 1987. Límites de placas y acortamientos recientes entre los paralelos 5°N y 8°N, Andes Colombianos. *Revista Geologica de Chile* 31, 95–100.
- Trenkamp, R., Kellogg, J.N., Freymueller, J.T., Mora, H., 2002. Wide plate margin deformation, southern Central America and northwestern South America, CASA GPS observations. *Journal of South America Earth Sciences* 15, 157–171.
- Vaca, S., Regnier, M., Bethoux, N., Alvarez, V., Pontoise, B., 2010. Sismicidad de la región de Manta (Ecuador): enjambre sísmico de Manta–2005 in *Geología y Geofísica Marina y Terrestre del Ecuador. Spec. Pub. INOCAR-IRD*, pp. 151–166.
- Virieux, J., Farra, V., Madariaga, R., 1988. Ray tracing in laterally heterogeneous media for earthquake location. *Journal of Geophysical Research* 93, 6585–6599.
- Waldhauser, F., Ellsworth, W.L., 2002. Fault structures and mechanics of the Hayward fault, California from double-difference earthquake locations. *Journal of Geophysical Research* 107 ESE 3–1–3–15.
- Winter, T., Avouac, J.P., Lavenue, A., 1993. Late Quaternary kinematics of the Pallatanga strike-slip fault (Central Ecuador) from topographic measurements of displaced morphologic features. *Geophysical Journal International* 115, 905–920.
- Witt, C., Bourgeois, J., Michaud, F., Ordóñez, M., Jiménez, N., Sosson, M., 2006. Development of the gulf of Guayaquil (Ecuador) during the Quaternary as an effect of the North Andean block tectonic escape. *Tectonics*. doi:10.1029/2004TC001723.
- Wortel, R., Cloetingh, S., 1981. On the origin of the Cocos–Nazca spreading center. *Geology* 9, 425–430.
- Wortel, R., 1984. Spatial and temporal variations in the Andean subduction zone. *Journal of the Geological Society* 141, 783–791.